Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, T_{2u} doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per `Docs/status/TOE-FACT-SHEET.md` should be re-calibrated against Math411-AddB §10.

The spin-c Dirac index on $\mathbb{CP}^2$: a corrected rank-dependent formula and the Picard classification of holomorphic line bundles

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We correct a degree-arithmetic error in an earlier internal computation (Math171 §3.3) of the spin-c Dirac index on the complex projective surface $\mathbb{CP}^2$. For a rank-r holomorphic vector bundle E with $c_1(E) = 0$ and $c_2(E) = \mu H^2$ (where $H \in H^2(\mathbb{CP}^2, \mathbb{Z})$ is the hyperplane class), the Hirzebruch–Riemann–Roch / spin-c Dirac index is

$$\mathrm{ind}(D^c_E) \; = \; r - \mu,$$

which evaluates to $16 - \mu$ in the rank-16 specialisation relevant to the SO(10) Standard-Model fermion sector. The general formula and its rank-16 specialisation are stated as a main theorem and an explicit corollary, respectively, to avoid conflating the two. Separately, we collect the standard fact $\mathrm{Pic}(\mathbb{CP}^2) \cong \mathbb{Z}$ as a remark on the discrete classification of holomorphic line bundles; this remark applies to the $\mathrm{U}(1)_\chi$ line-bundle classification on a $\mathbb{CP}^2$-type base and is NOT to be conflated with the rank-16 index formula above. The current canonical TECT realisation programme operates on the base $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ per Math305; the present results therefore form a foundational algebro-geometric background note rather than a direct closure of any present-canonical Pillar 4 sub-task.

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I. INTRODUCTION AND SCOPE

The Atiyah–Singer index theorem is one of the central tools of modern differential geometry and mathematical physics. Its spin-c variant extends the Dirac index formula to non-spin manifolds by twisting with a determinant line bundle. On $\mathbb{CP}^2$, a complex projective surface of central importance in algebraic geometry and gauge theory, the application of this formula requires careful attention to the degree-parity decomposition.

In an earlier internal computation (Math171 §3.3) the author wrote a formula of the form $\mathrm{ind}(D^c_E) = 16 - \mu$ for a holomorphic bundle E with $c_1(E) = 0$ and $c_2(E) = \mu H^2$. The present note clarifies that the correct general formula is rank-dependent, namely $\mathrm{ind}(D^c_E) = r - \mu$, and that the formula $16 - \mu$ is the specialisation to rank $r = 16$ (the rank relevant to the SO(10) chiral fermion sector targeted by the TECT Pillar 4 programme). We separate the index formula from the Picard classification of holomorphic line bundles to prevent the rank-1 and rank-16 contexts from being conflated.

The present note is a *background mathematical correction* rather than a TECT closure paper. The current canonical Pillar 4 realisation programme uses the base $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ per Math305, so the $\mathbb{CP}^2$ index formula derived here is applicable to earlier $\mathbb{CP}^2$-based versions of the Pillar 4 bundle programme but does not by itself close the current Σ_0-based realisation chain.

II. SETTING

A. Complex manifolds and spin-c structures

Recall $\mathbb{CP}^2 = (\mathbb{C}^3 \setminus \{0\})/\mathbb{C}^*$, a smooth real 4-manifold and a complex 2-manifold. Its tangent bundle $T\mathbb{CP}^2$ is a holomorphic vector bundle with first Chern class $c_1(T\mathbb{CP}^2) = 3H$, where H is the hyperplane class in $H^2(\mathbb{CP}^2, \mathbb{Z}) \cong \mathbb{Z}$.

The manifold $\mathbb{CP}^2$ is *not* spin since $w_2(T\mathbb{CP}^2) \neq 0$, but it admits a spin-c structure. The *canonical* spin-c structure has determinant line bundle L_{can} with $c_1(L_{\mathrm{can}}) = c_1(T\mathbb{CP}^2) = 3H$. Throughout this note we use the canonical spin-c structure.

B. Notation for bundles

We consider a holomorphic vector bundle E on $\mathbb{CP}^2$ of rank r, with first Chern class $c_1(E) = 0$ and second Chern class $c_2(E) = \mu H^2$ for an integer μ. The Chern character is

$$\mathrm{ch}(E) \;=\; r + c_1(E) + \tfrac{1}{2}(c_1(E)^2 - 2c_2(E)) + \cdots \;=\; r - \mu H^2 \quad (\text{mod } H^4 \text{ vanishes on } \mathbb{CP}^2),$$

where the $c_1(E)$ and $c_1(E)^2$ terms vanish under the $c_1(E) = 0$ hypothesis.

III. MAIN THEOREM (GENERAL RANK r)

Theorem 1 (Spin-c Dirac index on $\mathbb{CP}^2$, general rank)**.** *Let E be a holomorphic vector bundle on $\mathbb{CP}^2$ with $\mathrm{rk}(E) = r$, $c_1(E) = 0$, and $c_2(E) = \mu H^2$ for some integer $\mu \in \mathbb{Z}$ (where H is the hyperplane class generating $H^2(\mathbb{CP}^2, \mathbb{Z}) \cong \mathbb{Z}$ and the intersection $H^2 \in H^4(\mathbb{CP}^2, \mathbb{Z}) \cong \mathbb{Z}$ satisfies $\int_{\mathbb{CP}^2} H^2 = 1$). Let D_E^c denote the spin-c Dirac operator on $\mathbb{CP}^2$ coupled to E with the canonical spin-c structure $c_1(L_{\mathrm{can}}) = 3H$. Then*

$$\boxed{\mathrm{ind}(D_E^c) \;=\; r - \mu \;\in\; \mathbb{Z}.}$$

Proof. By the Atiyah–Singer index theorem in spin-c form, on a complex manifold of complex dimension 2 with the canonical spin-c structure, the index reduces to the Hirzebruch–Riemann–Roch holomorphic Euler characteristic:

$$\mathrm{ind}(D_E^c) \;=\; \chi(\mathbb{CP}^2, E) \;=\; \int_{\mathbb{CP}^2} \mathrm{td}(T\mathbb{CP}^2) \wedge \mathrm{ch}(E).$$

The Todd class of $T\mathbb{CP}^2$ is

$$\mathrm{td}(T\mathbb{CP}^2) \;=\; 1 \;+\; \tfrac{3}{2}H \;+\; H^2.$$

The Chern character of E, under the hypotheses $c_1(E) = 0$ and $c_2(E) = \mu H^2$, has degree-graded components

$$[\mathrm{ch}(E)]_{(0)} = r, \qquad [\mathrm{ch}(E)]_{(2)} = 0, \qquad [\mathrm{ch}(E)]_{(4)} = -\mu H^2.$$

The degree-4 component of the wedge product $\mathrm{td}(T\mathbb{CP}^2) \wedge \mathrm{ch}(E)$ is

$$\begin{aligned}[\mathrm{td} \wedge \mathrm{ch}]_{(4)} &= [\mathrm{td}]_{(4)} \cdot [\mathrm{ch}]_{(0)} + [\mathrm{td}]_{(2)} \cdot [\mathrm{ch}]_{(2)} + [\mathrm{td}]_{(0)} \cdot [\mathrm{ch}]_{(4)} \\ &= H^2 \cdot r + \tfrac{3}{2}H \cdot 0 + 1 \cdot (-\mu H^2) \\ &= (r - \mu)\, H^2.\end{aligned}$$

With $\int_{\mathbb{CP}^2} H^2 = 1$ this integrates to $r - \mu$. $\square$

IV. COROLLARY: RANK-16 SPECIALISATION

Corollary 2 (Rank-16 specialisation)**.** *For the rank-16 specialisation $r = 16$ relevant to the SO(10) chiral-fermion sector of the TECT Pillar 4 programme,*

$$\mathrm{ind}(D_E^c) \;=\; 16 - \mu.$$

The rank-16 specialisation reproduces the formula stated in the earlier Math171 §3.3 derivation; the present note clarifies that the formula is rank-dependent and that the original "$16 - \mu$" wording is the $r = 16$ corollary, not a general theorem.

V. REMARK: PICARD CLASSIFICATION OF HOLOMORPHIC LINE BUNDLES ON $\mathbb{CP}^2$ (SEPARATE FROM THE INDEX THEOREM)

Remark 3 (Picard classification, separate from Theorem 1)*.* Independently of Theorem 1, the Picard group of $\mathbb{CP}^2$ is

$$\mathrm{Pic}(\mathbb{CP}^2) \;\cong\; \mathbb{Z}, \qquad \mathrm{Pic}^0(\mathbb{CP}^2) = 0.$$

This follows from the standard exponential sequence and from $H^1(\mathbb{CP}^2, \mathcal{O}) = 0$ (Kodaira vanishing on $\mathbb{CP}^2$). Consequently, every holomorphic line bundle on $\mathbb{CP}^2$ is uniquely determined by an integer first Chern class $k \in \mathbb{Z}$, with no continuous moduli. If a TECT realisation is formulated on a $\mathbb{CP}^2$-type base and the $\mathrm{U}(1)_\chi$ sector is represented by a holomorphic line bundle, that line bundle is classified discretely by an integer first Chern class k_χ and has no Pic^0-modulus.

The Picard classification (Remark 3) and the spin-c index formula (Theorem 1) are mathematically independent statements. They concern different bundle ranks and different invariants; this note keeps them separated to prevent the rank-1 Picard context from being conflated with the rank-r index calculation.

VI. DISCUSSION

The corrected rank-dependent formula $\mathrm{ind}(D_E^c) = r - \mu$ removes the degree-arithmetic ambiguity in the earlier Math171 §3.3 derivation. The rank-16 specialisation $16 - \mu$ is preserved as Corollary 2; the rank-1 specialisation $1 - \mu$ applies, e.g., to a hypothetical line-bundle context with $c_2 = \mu H^2$ (note that for an honest holomorphic line bundle with $c_1 = 0$ the second Chern class c_2 vanishes, so the rank-1 specialisation is non-trivial only for the abstract index calculation, not for an honest line bundle in the geometric sense).

Per the current canonical TECT bookkeeping (Math305 plus Math302 proof-rigor / realisation carve-out), the canonical Pillar 4 base is $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ rather than $\mathbb{CP}^2$. The index formula on Σ_0 requires a separate calculation (different Todd class) and is not derived here. The present note therefore serves as a background algebro-geometric clarification of the earlier $\mathbb{CP}^2$ programme rather than as a current canonical-tier closure.

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